UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	NOVEMBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of four (4) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

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QUESTION 1

a) Solve the inequality $\frac{6}{5-3x} \geq 3$. Express the answer in interval notation.
(5 marks)

b) If $Z_1 = -1 + 2\mathrm{i}$ and $Z_2 = 4 - 3\mathrm{i}$, state $\frac{\overline{Z_2}}{\overline{Z_1}\mathrm{i}^{19}}$ in standard form $a + bi$.
(5 marks)

QUESTION 2

a) Solve the inequality $\frac{2x-5}{x-2} \leq 1$ and express the solution in real number line.
(5 marks)

b) Given the equation $2(y-3)^2 + 32 = 16x$, find the standard form of the parabola, the vertex and the equation of directrix. Hence, sketch the parabola with correct vertex and directrix:
(5 marks)

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QUESTION 3

a) Solve the following systems of equations simultaneously:
i)

(0 marks)

b) Graph the following system of inequalities below and shade the feasible region.

(5 marks)

c) Solve the trigonometric equation $\cot^2\theta + \operatorname{cosec}\theta = 5$ for $0^\circ \le \theta \le 360^\circ$
(5 marks)

QUESTION 4

a) Given $f(x) = \frac{x+2}{x-1}$ and $g(x) = 1 + 4x$. Determine

i) the domain of $f(x)$ and $g(x)$.
(2 marks)

ii) $(f+g)(x)$.
(3 marks)

iii) $f^{-1}(5)$.
(4 marks)

b) Use long division to show that x - 1 is not a factor of $P(x) = 2x^3 - x^2 + 1$. Hence, state the quotient and remainder of $P(x)$.
(5 marks)

c) Solve the equation $\log_3 x + 4\log_x 3 = 5$.
(6 marks)

d) Figure 1 shows a graph with a function $f(x)$. Use the transformation of graphs to illustrate the function $g(x) = -f(x+4) + 5$. Hence, state the y-intercept (if any).
(5 marks)

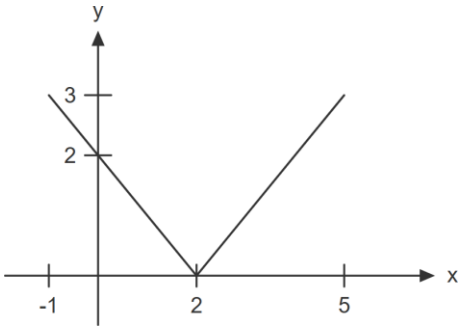


Figure 1

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

1. $\sin^2 A + \cos^2 A = 1$
2. $\sin(A + B) = \sin A\cos B + \cos A\sin B$
3. $\cos(A + B) = \cos A\cos B - \sin A\sin B$
4. $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A\tan B}$
5. $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
6. $\sin 2A = 2\sin A\cos A$
7. $\tan^2 A + 1 = \sec^2 A$
8. $\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$
9. Arc, $s = r\theta$
10. Area of sector $= \frac{1}{2}r^2\theta$
11. $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
12. $a^2 = b^2 + c^2 - 2bc\cos A$
13. Area $= \frac{1}{2}ab\sin C$
14. Area $= \sqrt{s(s-a)(s-b)(s-c)}$

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